MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 12AMP19

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus, aluminized screen

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb1	Eb2	Eb3	ed	Rg	Ehk	Eb3/Eb2	Mean deflection plate
Unit:	V	Vdc	Vdc	Vdc	Vdc (Note 5)	Vdc	v	Meg	Vdc	Ratio (Note 2)	V
Maximum:	6.9	0	200	3,000	8,500	15,000	1,500	2.0	+180	2.5	---
Minimum	5.7	-100	---	---	---	---	---	---	-180	---	---
TEST CONDITIONS:	6.3	Adj	Adj (Note 3)	Focus	Adj (Note 4)	9,200	---	---	---	---	5,300

GENERAL:

Qualification - Required

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
5248	Deflection-factor Uniformity	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	Cathode to all	Ck	---	6	pF
			Grid 1 to all	Cg1	---	12	pF
			D1 to D2	C1D2	---	5	pF
			D3 to D4	C3D4	---	4	pF
			D1 to all	CD1	---	13.5	pF
			D2 to all	CD2	---	13.5	pF
			D3 to all	CD3	---	10	pF
			D4 to all	CD4	---	10	pF
1141	Pressure (implosion)	-		---	---	---	---
5111	Vibration	-		---	---	1.5	mm
1027	Temperature cycling (nonoperating)	10	5 cycles, 0° to 50°C	---	---	---	---
(B) 1216	Base material insulating quality	-	Zone 5 minimum	---	---	---	---
	Quality conformance inspection, part 1						
5201	Electrode current (anode No. 1)	9	Ib3 = 55 μAdc	Ib1	-5	+5	μAdc
5201	Electrode current (cathode)	9	Ib3 = 55 μAdc	Ik	---	125	μAdc
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	8	Ib3 = 5 μAdc	---	---	---	---
5106	Bulb, screen, and faceplate quality	-	Dead spots	Diameter	---	0.040	Inch
			Number	Per tube	---	12	Spots
5241	Grid-cutoff voltage	3,4	+40 ≤ Ec2 ≤ -120 Vdc	Eco	0	-11.5	Vdc
5223	Modulation (cathode driven negative from cutoff voltage)	9	Ec2 determined from Eco test: Ib3 = 55 μA	Ek	---	11.5	Vdc
---	Astigmatism voltage	6		---	0	+75	Vdc
5231	Spot position (electrostatic deflection)	-		---	---	19	mm
5231	Spot displacement (leakage)	-		---	---	12.7	mm
---	Cathode leakage	-	Ec1 biased to cutoff	---	---	5	μAdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	5		---	---	---	---
---	Focus correction voltage	7	Deflection = 5 inches	---	0	+370	Vdc
---	Total astigmatism correction voltage	7	1D2; deflection = 5 inches	---	0	+150	Vdc

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MINIMUM	LIMITS MAXIMUM	UNIT
	Quality conformance inspection, part 1 -Continued						
---	Critical anode voltage ratio	-		Eb3 Eb2	---	1.85	---
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5101	Base alinement (electrostatic types)	-	+1D2, pin No. 1	---	---	---	---
5101	Side terminal alinement (electrostatic types)	-	+1D2	---	---	---	---
5101	Side terminal and base alinement	-	Pin No. 1	---	---	---	---
5101	Neck and bulb alinement (electrostatic types)	-		---	---	4.312	Inches
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
5216	Stray light emission (conventional types)	-	Eb2 = 8,500 Vdc; Eb3 = 15,000 Vdc	---	---	---	---
5221	Screen (P19)	1		cBf D0.1 D1.0 D1.0-D10.0	575 410 335 75	--- --- --- 120	cB cB cB cB
5246	Focusing voltage (zero-bias)	-		Eb1	1,450	1,750	Vdc
1101	Secureness of base, cap, or insert	-		---	---	---	---
(B) 5101	Angle between traces	-		---	89	91	Degrees
(B) 5251	Grid No. 2 leakage current	-		---	---	---	---
(B) 5226	Line width A (electrostatic deflection)	-	Ib3 = 2 μAdc	---	---	0.63	mm
(B) 5226	Line width B (electrostatic deflection)	-	Ib3 = 2 μAdc	---	---	0.84	mm
(B) 5246	Focusing voltage at cutoff	-		Eb1	1,450	1,750	Vdc
(B) 5248	Deflection factor	-	1D2	DF	105	117	Vdc/in.
(B) 5248	Deflection factor	-	3D4	DF	105	117	Vdc/in.
(B) 5251	Heater-cathode leakage	-		Ihk	---	-5.0	μAdc
(B) 1111	Base pin solder depth	-		---	---	---	---
(B) 1105	Permanence of marking	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3						
---	Life test	-	Group C: Ib3 = 2 μAdc; t = 500 hours (min)	---	---	---	---
---	Life-test end points:						
5223	Modulation	-	Ib3 = 35 μAdc	Ek	---	11.8	Vdc
5226	Line width A	-	Ib3 = 2 μA	---	---	0.71	mm
5226	Line width B	-	Ib3 = 2 μA	---	---	0.96	mm

NOTES:

1. The phosphorescent light intensity observed t seconds after completion of the application of the nth raster to the tube is Dt(n), (for example, D1 (5) is the phosphorescent light intensity 1 second after the 5th excitation raster). cBf shall be the fluourescent light output under continuous raster excitation. The screen characteristics shall be measured under the following conditions:

   (a) Raster size (focused beam): 7.1 cm by 7.1 cm giving a total area of 50 cm^2. Tube shall be defocused to a spot approximately 0.25 cm in diameter.
   (b) Beam current: 2 μA at tube operating voltage of Eb3 = 9.200 volts.
   (c) Duration of raster: 1/60 second. The raster is formed by means of linear scanning frequency of 12 kHz and the vertical scanning frequency of 60 Hz. The raster is repeated at 1 second intervals.
   (d) Distance of raster from axis of the calibrated 931 VA multiplier tube: 30 cm.
   (e) White light: Sufficient to de-excite the phosphor before build up measurements so that the results are within 3 cB of those that would have been obtained after total de-excitation. The raster shall be repeated 60 times per second. After the tenth raster, excitation shall be removed. The light output 0.1 second (D0.1), 1 second (D1.0), and 10 seconds (D10.0) after excitation is removed shall satisfy the specified limits. Before performing the above measurements, illuminate screen with a 250 watt projection lamp for approximately 1 minute, then wait 30 seconds after lamp is turned off. Centibels may be converted to millifoot lamberts by the use of the following equation:

$$mftL = 1.77 \times 10^{-3} \times 10^{\frac{cB}{100}}$$

2. Tube shall operate acceptably within minimum usable face diameter with Eb3 Eb2 ratios up to 2.0. The critical anode voltage ratio shall be that Eb3 Eb2 which causes deflection plate shadow tangent to but not encroaching upon either extreme of the minimum useful screen diameter along the 1D2 axis.

3. Grid No. 2 voltage will be adjusted so that cathode modulation and grid cutoff voltage requirements are met; this grid No. 2 voltage shall fall within the specified range.

4. Grid cutoff voltage shall be determined by cutoff of a focused, undeflected spot, with the tube placed in a light tight enclosure; the observer shall have accomodated his eyes to viewing the tube's screen under these conditions for at least 2 minutes. For this test, Ec2 refers to the voltage between Ec2 and Ec1.

5. Anode No. 2 and astigmatism elements shown on figure 1 are to be connected externally. All other references to anode No. 2 in this specification are concerned with the combination of these elements.

6. Astigmatism voltage shall be defined as the potential difference between A2 and the mean deflection plate potential (measuring A2 with respect to mean deflection plate voltage). Operating the tube under the specified test conditions, a low intensity spot is displayed at the geometric center of the tube face. The focus (A1) voltage and astigmatism voltage are then adjusted for optimum spot size and roundness. The astigmatism voltage shall fall within the limits specified and subsequent electrical tests shall be made with the tube operating at this astigmatism voltage setting.

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NOTES: -Continued

7. Focus and astigmatism correction voltages are determined in the following manner: Operate the tube as specified in note 6 and measure focus voltage and mean deflection plate voltage. Maintaining the same mean deflection plate voltage, move the spot successively to four points ±5 inches (127 mm) from the geometrical center of the tube face on the 1D2 and 3D4 axes. At each point on the 3D4 axis, adjust focus voltage for optimum spot size and roundness. Measure focus voltage at each point. At each of the points on the 1D2 axis adjust both focus and astigmatism voltages for optimum spot size and roundness. Measure focus and astigmatism voltages at each point. The variation in focus voltage readings at the four deflected positions shall not exceed 150 volts. The difference between the deflected and undeflected focus voltage readings is called the focus correction voltage and shall be within the limits specified at all four points. The variation in astigmatism voltage readings at the 1D2 deflected positions shall not exceed 50 volts. The difference between the deflected and undeflected astigmatism voltage reading is called the astigmatism correction voltage. The difference between the deflected astigmatism voltage reading and the mean deflection plate voltage is called the total astigmatism correction voltage and shall be within the limits specified.

8. This test to be performed at the conclusion of the holding period.

9. The P19 screen can be permanently damaged if the current density is permitted to rise too high. To prevent burning, minimum beam current densities should be employed. Tests requiring high beam currents may be made by measuring beam current at the deflection plates. To make this measurement, deflect a 1 by 1 inch low intensity raster in the 1D2 direction until the raster cannot be seen on the screen. Beam current is then measured at D1 or D2 with Eb2 = Eb3 = 5,300 Vdc.

10. During qualification testing, this test shall be performed prior to all other tests. In addition to meeting all electrical tests, there shall be no evidence of cracks or other glass defects following the temperature cycling tests.

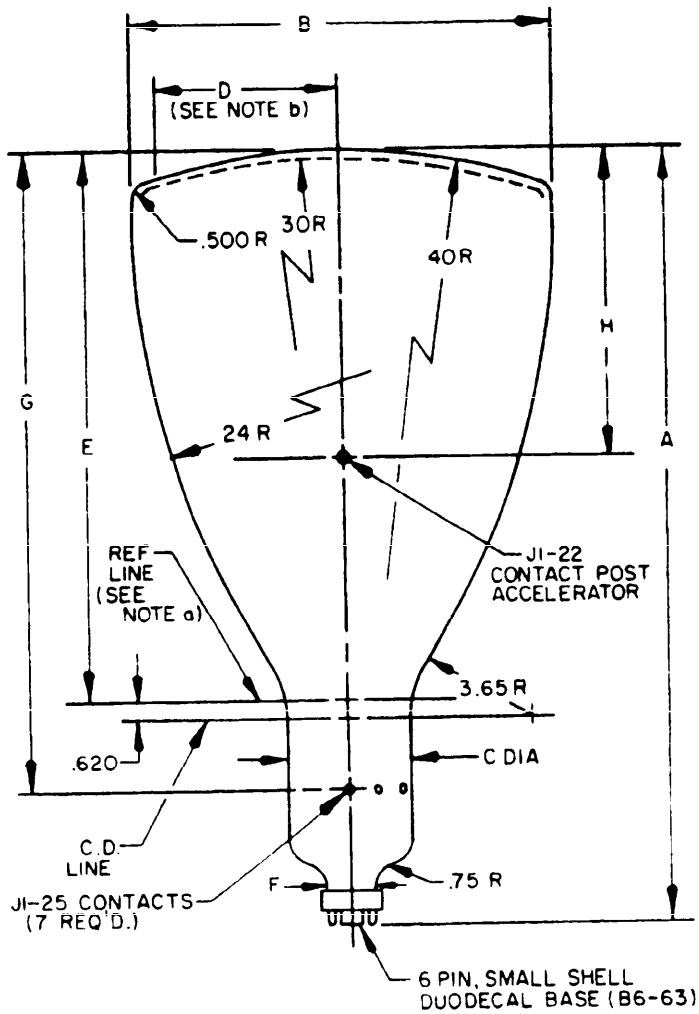
User activities:
Navy - AS, OS, MC, CG, SH

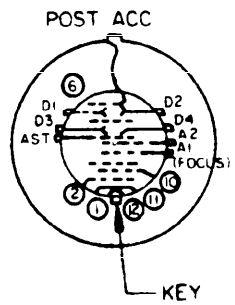
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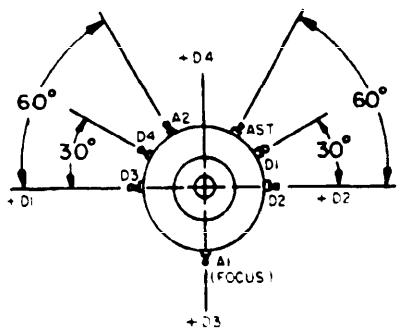
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BASING DIAGRAM

PIN NO.	CONNECTION
1	HEATER
2	GRID NO. 1
6	NO CONNECTION
10	GRID NO. 2
11	CATHODE
12	HEATER



BOTTOM VIEW

FIGURE 1. Outline drawing of electron tube type 12AMP19.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
Base: B6-63		
Quality conformance inspection, part 2		
A	22.75(577.85)	23.50(596.90)
B	12.37(314.20)	12.50(317.50)
C	3.68 (93.47)	3.82 (97.03)
D	5.50(139.70)	
E	15.70(398.78)	16.20(411.48)
F	1.37 (34.80)	1.51 (38.35)
G	18.25(463.55)	19.00(482.60)
H	8.50(215.90)	9.00(228.60)

NOTES:
a. Defined as that point where a 3.857 ± .003(97.97 ± .08 mm) ID ring gage comes to rest against bulb cone.
b. Useful screen radius.

FIGURE 1. Outline drawing of electron tube type 12AMP19 - Continued.

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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